

GitHub-State Comparison and Continuum-Foundations Synchronization Release for HAOS-IIP

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Abstract

This paper freezes the repository-comparison state of HAOS-IIP immediately before the next public synchronization push. The live GitHub branch remains at commit `cf5d48f`, which is the frozen geometry-emergence release 47.2. The local repository, however, now contains a coherent continuum-facing tranche that has not yet been published: strengthened scalar operator-convergence controls, harmonic / coexact vector-sector separation work, the bounded HAOS-to-harmonic theorem ladder from T1 through T8, the first active-sector transport machinery for the derived coexact physical branch, and two already prepared synthesis papers (48.1 and 49.1). This paper does not claim new physics beyond those artifacts. Its role is narrower and operationally important. It records exactly what is live on GitHub, exactly what exists locally above that live state, what remains open, what belongs in the next synchronized release, and what should still be excluded because it is unfinished or unrelated. The result is release 50.1: a bounded synchronization paper that closes the comparison gap between public GitHub state and local repository state without overstating unfinished disorder-threshold work.

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1 Scope

The purpose of this paper is not to introduce a new theorem or a new numerical phase contract. Its purpose is to freeze the comparison between:

- the live GitHub repository state;
- the current local HAOS-IIP worktree;
- the intended release scope for the next public synchronization push.

This is needed because the local repository has materially outrun the public GitHub branch. A clean synchronization release should say so explicitly rather than silently mixing comparison, numerics, foundations, and publish logistics into one undocumented push.

2 Live GitHub Baseline

At the time of comparison, the live remote branch is:

- repository: `tomislavrupic/HAOS-IIP`;
- branch: `main`;
- live commit: `cf5d48fb2178b92879a98dcdf0d52a4ec90ce60e`;
- live release headline: “Freeze geometry_emergence release 47.2”.

That means the public GitHub branch still reflects the geometry-emergence tranche and does not yet contain the later continuum and derivation stack.

3 Local Delta Above the Live GitHub State

Relative to live GitHub 47.2, the local repository contains one coherent research-and-release tranche. It can be organized into four blocks.

3.1 Continuum-scaling and operator updates

The scalar branch is no longer limited to the older clean interior cubic check. The local repository now includes:

- perturbed point-set scalar controls;
- reflected boundary-condition controls;
- a strengthened scalar convergence note and updated result payload;

- harmonic / coexact sector-aware transverse comparison;
- full-window candidate and direct gap scans for the vector branch.

These artifacts collectively justify the bounded scalar statement that later became Paper 48.1, while keeping the vector branch bounded by a real harmonic-floor obstruction rather than rhetorical optimism.

3.2 HAOS-to-harmonic theorem ladder

The largest conceptual change above live GitHub is the new foundations stack. The local repository now contains a bounded theorem ladder:

$$T1 \rightarrow T2 \rightarrow T4 \rightarrow T5 \rightarrow T6 \rightarrow T7 \rightarrow T8,$$

with the scalar toy bridge and locality cleanup notes filling important gaps.

This does not prove that HAOS explains all of harmonic mathematics. It does something narrower and operationally useful:

it shows how much Laplacian, Hodge, sector-decomposition, physical-sector, and refinement-stability structure can be derived from recoverable coherence under interaction before the argument becomes assumption rather than theorem.

That bounded ladder is not present on live GitHub.

3.3 Active-sector transport machinery

The local repository also contains the first explicit T8 execution tools. The derived physical branch is no longer compared only by separate spectra. Instead, restricted coexact modes are carried into a common probe space by explicit comparison maps J_n , and then compared by:

- transported overlaps;
- principal-angle cosines;
- scaled-eigenvalue drift.

This is a genuine structural change in the repository because it moves the vector-sector refinement question from intuition to an explicit transported-branch identity test.

3.4 Release papers already prepared locally

Two synthesis papers exist locally but are not yet live on GitHub:

1. 48.1 Scalar Continuum Control, Harmonic-Coexact Sector Separation, and a Bounded Roadmap Toward Continuum Physics on HAOS-IIP;
2. 49.1 From Recoverable Coherence to Harmonic Operator Structure: A Bounded Derivation Program, Physical-Sector Restriction, and Continuum-Stability Roadmap for HAOS-IIP.

This paper becomes the third release-layer artifact in the tranche:

3. 50.1 GitHub-State Comparison and Continuum-Foundations Synchronization Release for HAOS-IIP.

4 Detailed Comparison Map

Table 1 compresses the comparison between the live GitHub state and the local repository state.

Table 1: Repository delta from live GitHub 47.2 to the local synchronization tranche.

Area	Live GitHub state	Local state above GitHub
Scalar continuum	Geometry-emergence branch only	Strengthened scalar convergence controls with perturbed point sets, boundary conditions, updated note, and result payload
Vector-sector comparison	No bounded harmonic / coexact comparison tranche live	Harmonic/coexact split comparison, candidate-window scans, and direct harmonic-to-coexact gap scans
Foundations derivation	No public HAOS-to-harmonic theorem ladder	Bounded executed notes for T1--T8 plus scalar-to-graded bridge and F1 cleanup
Physical-sector transport	No public active-sector transport map	Explicit J_n transport runner, bounded transport note, mild-disorder extension through $n = 28$
Paper spine	Ends at 47.2	Adds prepared papers 48.1, 49.1, and the present 50.1 synchronization paper
Execution plan	No public bounded work-package note for the continuum tranche	Explicit execution-program note ordering F1--F4, N1--N3, C1--C4

5 What Is Stronger Now

Three things are materially stronger than the public GitHub branch suggests.

5.1 The scalar authority boundary

The scalar branch now has a real bounded continuum-facing contract in the local kernel regime. This is not a grand continuum claim, but it is stronger than the older public state.

5.2 The foundations authority boundary

The repository no longer only *uses* harmonic mathematics. It now contains a bounded argument for why harmonic operator structure is a natural normal form of recoverability on graded relational substrates under explicit HAOS-aligned assumptions.

5.3 The vector diagnostic boundary

The vector branch is still open, but it is open in a sharper way. The active coexact branch can now be tracked explicitly under refinement, and the clean baseline weakness can be separated from the

mild-disorder and defect-supported cases rather than being left as an ambiguous nonconvergence statement.

6 What Is Still Open

This synchronization paper is only honest if it states the remaining open items clearly.

6.1 Foundational open items

The following are still not closed:

- the incidence-normalization caveat (F2);
- channel completeness and the no-extra-bridge rule (F3);
- the Hilbert-complex upgrade of the finite-dimensional sector theorem (F4).

6.2 Vector-sector open items

The vector branch still lacks:

- a frozen disorder-threshold result bundle;
- full active-branch refinement closure in the clean baseline;
- universality, effective-law, and geometry / response closure on the same transported branch.

The interrupted disorder-threshold work therefore remains active work, not release-level evidence.

7 Release Scope for 50.1

The intended release scope for 50.1 is:

- HAOS-IIP numerics associated with the continuum / derivation / transport tranche;
- new foundations notes;
- experiment notes, result bundles, and plots already cleanly stamped in the repository;
- prepared papers 48.1 and 49.1;
- the present synchronization paper and release-index updates.

The release should explicitly exclude:

- unrelated image assets;
- any unfinished disorder-threshold outputs that were not cleanly frozen into the standard result path.

8 Recommended Publish Action

The correct next public action is not to pretend that GitHub is already current. It is to synchronize the repository honestly by publishing one coherent tranche that moves public state from 47.2 to the present bounded continuum / derivation / transport state.

That synchronized push should make the following true simultaneously:

1. GitHub contains the scalar continuum and vector-sector boundary work of Paper 48.1;
2. GitHub contains the bounded HAOS-to-harmonic derivation stack of Paper 49.1;
3. GitHub contains the active-sector transport tools and execution-program note;
4. GitHub contains the explicit repository-comparison record of Paper 50.1;
5. GitHub still does *not* overclaim a finished vector-sector continuum closure.

9 Conclusion

The live GitHub repository is still frozen at a geometry-emergence release. The local repository is already farther along. It contains a coherent continuum-facing tranche with stronger scalar evidence, a bounded HAOS-to-harmonic derivation stack, and the first explicit active-sector refinement transport machinery. That tranche is real and should be published. At the same time, the vector branch remains numerically unfinished, and the disorder-threshold question should remain outside the frozen claim set until a clean result bundle exists. Release 50.1 is therefore not a new physics claim. It is the synchronization document that makes the public state match the actual bounded local state.